Break-even inflation in Panels B to D of Figure 11.5 is represented by the distance between the TIPS yield (sold line) and the Treasury yield (dashed line). Taking out the financial crisis in 2008 and 2009, break-even inflation is fairly stable. For example, for the ten-year maturity TIPS and Treasury bonds in Panel C, break-even inflation was 2.44% from July 2004 to December 2007, and from January 2010 to December 2012, it was 1.79%. For the twenty-year maturity bonds in Panel D, the corresponding numbers for the two periods are 2.66% and 2.35%.

TIPS yields are set by the market relative to Treasuries yields, for reasons that are not entirely clear, such that break-even inflation is fairly stable. During and after the financial crisis, the nominal yield on long-term bonds was pushed down partly by quantitative easing and other nonconventional monetary policies (see chapter 9) and partly as a flight-to-quality response by investors seeking the safety of Treasuries versus risky European and other sovereign debt. Except for the 2008 to 2009 period, the market prices TIPS yields at an approximately constant discount to Treasuries. Since Treasury yields are very low, this leads to negative real yields.

Break-even inflation, or inflation compensation, can be decomposed into two terms:

$$\text{Break-Even Inflation} = \text{Expected Inflation} + \text{Risk Premium.} \tag{11.2}$$

If the risk premium is constant, then break-even inflation moves one-for-one with expected inflation. But since risk premiums vary over time, break-even inflation cannot be used directly as a measure of future expected inflation. When the variation in the inflation risk premium is small, changes in TIPS yields relative to the benchmark Treasury curves are statements of the market's expectations about future inflation.[13] Inflation risk premiums, however, change. Ang, Bekaert, and Wei (2008) show they were especially large during the Great Inflation of the 1960s and 1970s, during the late 1980s when the economy was booming, and during the mid-1990s (see also chapter 9 on fixed income).

Using the decomposition of break-even inflation in equation (11.2), we can interpret the negative break-even inflation rates during the financial crisis as the market pricing in negative future inflation, or negative risk premiums, or both. Can we say which it is?

Inflation surveys turn out to be one of the best methods of forecasting inflation and beat many economic and term structure models, as Ang, Bekaert, and Wei (2007) show. Inflation expectations have been fairly stable for the sample considered in Figure 11.5, except in the financial crisis. At December 2008, the median

[13] Fleckenstein, Longstaff, and Lustig (2010) argue that TIPS and Treasuries are mispriced relative to each other with TIPS being cheap and Treasuries being expensive. They claim this is the "largest arbitrage ever documented in the financial economics literature."